The muscular system has three different types of muscles

- Cardiac muscle
- Skeletal muscle
- Smooth muscle

Cardiac muscles are striated muscles only found in the heart. Cardiac muscles works involuntarily without the control of a person.The cells found in cardiac muscle are the myocardiocytes which contract the myocardium.

Skeletal muscle is striated and voluntary, that enables the skeletal structures to move. These muscles are also controlled by the nervous system and the majority of skeletal muscles are attached to the bones of the body with tendons to enable body movement.

Smooth muscles are non striated and involuntary muscles. Muscles that control the movements and actions of the internal organs and systems of the body like peristalsis.

Role of muscular system

- **Voluntary movement** with the skeletal muscles
- **Involuntary actions** with the smooth muscles
- **Involuntary contractions and relaxation** of the heart with the cardiac muscles

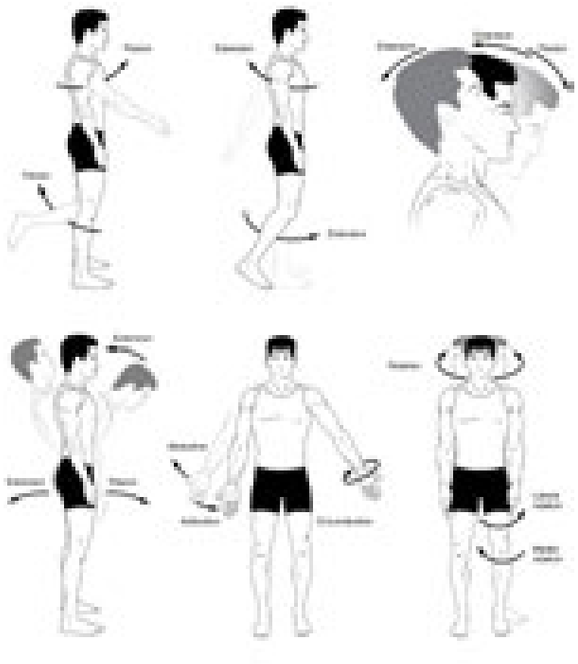
Abduction: Movement away from the centre of the body.

Adduction: Movement towards the middle of the body

Flexion: Bending movement of a joint or muscle

Hyperflexion: The flexion of a joint that is beyond what it normally should do, for example, can occur with a traumatic car crash when the head is forced to the chest, or flexed, beyond what it normally should do.

Extension: Normal straightening movement of a joint or muscle.

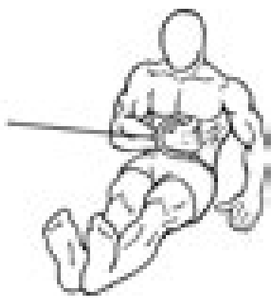


The Movements of Muscles

Hyperextension: Extension of a joint is beyond what it normally should do, for example, can occur with a traumatic car crash when the head is forced backwards, or extended, beyond what it normally should do.

Rotation: Circular movement of a joint or muscle that allows the body part to move in a circular manner.

External rotation: Muscular and joint movement that entails both circular movement and also movement away from the center of the body.



Rotating the arm closer to the body is internal rotation

Internal rotation: Muscular and joint movement that entails both circular movement and also movement towards the center of the body.

Unit_1_Human_physiology_Gen_Nur.indd 12 09-03-2020 13:03:03